CISCO PACKET TRACER SIMULATION

Experiment 3: Installing and Configuring a Basic Network

AIM

To install Cisco Packet Tracer, build a simple network with two computers linked through a switch, assign IP addresses, and check whether the computers can communicate.

REQUIREMENTS

A computer or laptop, an internet connection, Cisco Packet Tracer software, and an account for Cisco Networking Academy or Skills for All.

INTRODUCTION

Cisco Packet Tracer is a network simulation application that allows users to plan, build, and test computer networks virtually. It includes simulated devices such as PCs, switches, routers, and network cables, so configurations can be practised without needing physical equipment.

INSTALLATION PROCEDURE

Step 1: Visit the Cisco Networking Academy or Skills for All website and log in to your account.

Step 2: Open the Packet Tracer download page and choose the installation file that matches your operating system.

Step 3: Once the file has downloaded, find it on your computer and double-click it to launch the installer.

Step 4: Review the licence terms, accept the agreement, and select Next.

Step 5: Choose the folder where the application should be installed. Keep the default location if no change is needed.

Step 6: Choose the shortcuts you want to create, then click Next.

Step 7: Select Install and wait for the installation process to finish.

Step 8: Click Finish, then launch Cisco Packet Tracer.

Step 9: Sign in with the required Cisco account. The software is ready for use.

PROCEDURE FOR BASIC NETWORK SETUP

1. Start Cisco Packet Tracer.

2. Under End Devices, drag two PCs into the workspace.

3. From Network Devices → Switches, add one switch to the workspace.

4. Choose a Copper Straight-Through cable and connect PC0 to a port on the switch.

5. Use another Copper Straight-Through cable to connect PC1 to a different switch port.

6. Open PC0, choose Desktop → IP Configuration, and set the IP address to 192.168.1.10 with subnet mask 255.255.255.0.

7. Open PC1, choose Desktop → IP Configuration, and set the IP address to 192.168.1.11 with subnet mask 255.255.255.0.

8. On PC0, open Desktop → Command Prompt and enter the command shown below to test the connection.

```
ping 192.168.1.11
```

NETWORK CONFIGURATION

Device	IP address	Subnet mask
PC0	192.168.1.10	255.255.255.0
PC1	192.168.1.11	255.255.255.0

RESULT

Cisco Packet Tracer was installed, and a basic local area network was assembled using two PCs and a switch. The computers were assigned the specified IP addresses, and communication was checked by sending a ping from PC0 to PC1.